

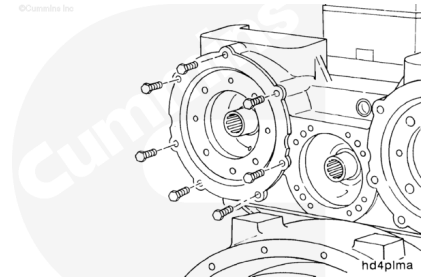
Trainings ▾

Remove

Note : Some engines will **not** have an outer hydraulic drive. These engines require a cover plate and a non-splined shaft, but do **not** use a hydraulic gear.

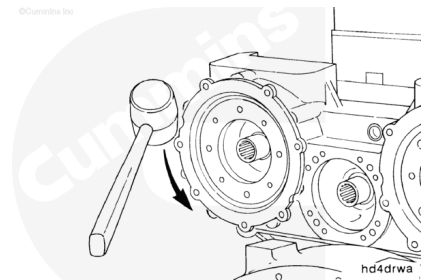
If a cover plate is used, remove the cover plate and non-splined shaft.

Remove the seven capscrews from the hydraulic drive.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Care must be taken to avoid possible breakage of the capscrew mounting flanges.

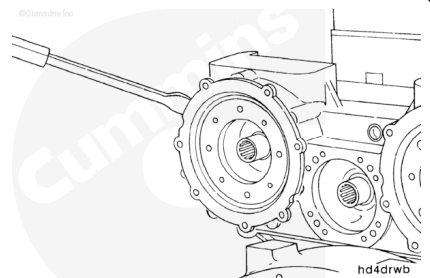
The outboard hydraulic pump supports **must** be rotated to remove them from the housing.

Use a mallet, and carefully tap the side of the support.

Remove the outboard hydraulic pump supports with a pry bar.

Remove the gaskets.

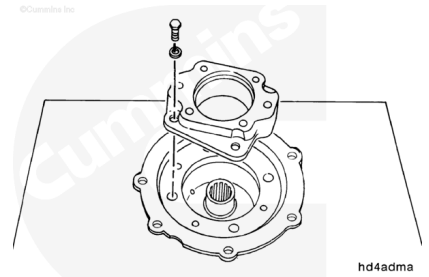
Discard the gaskets.



Disassemble

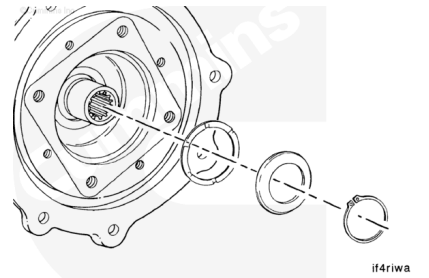
Remove the hydraulic pump adapter and/or cover plate.

Remove and discard the gasket.

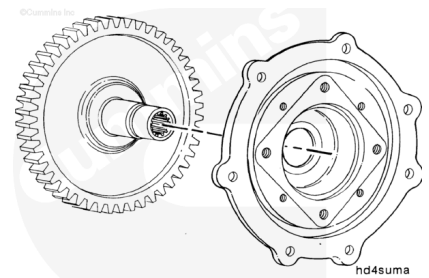


Remove the tapered retaining ring, bearing spacer, and thrust bearing.

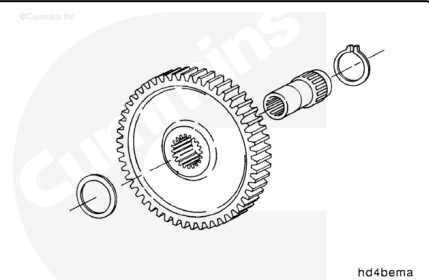
Check the bearing spacer for wear against the shaft shoulder.



Remove the hydraulic pump support from the splined drive shaft.

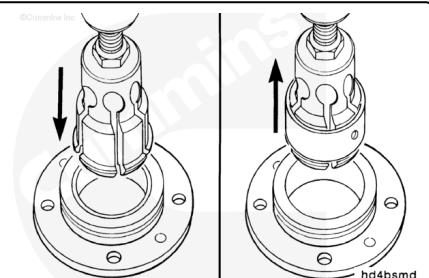


Remove the thrust bearing, gear, and non-tapered retaining ring from the drive shaft.



Note : Remove the bushing only if it is not within specifications.

Remove the bushing with a blind bushing puller. Refer to Procedure 009-037 (</qs3/pubsys2/xml/en/procedures/20/20-009-037.html>).



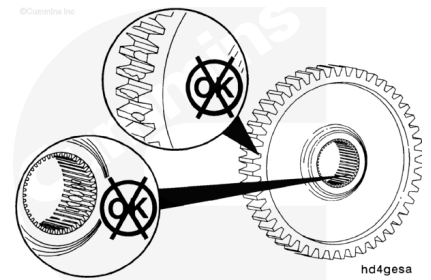
Clean and Inspect for Reuse

Clean the gear.

Inspect the idler gear for chipped, broken or cracked teeth.

Check the internal splines for wear.

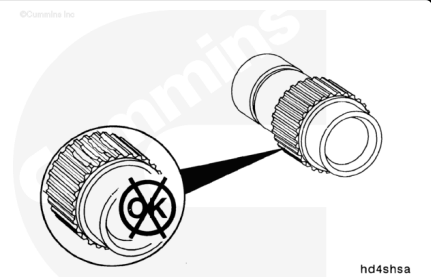
If the gear is damaged, it **must** be replaced.



Clean the hydraulic pump drive shaft.

Check the hydraulic pump drive shaft external splines for wear.

If the splines are excessively worn, the hydraulic pump drive shaft **must** be replaced.

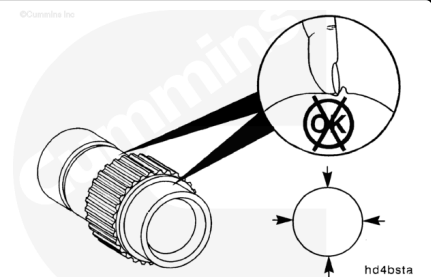


Inspect the bushing diameters on the shaft for wear.

If a fingernail will catch on scratches or grooves, the shaft **must** be replaced.

Measure the diameter of the shaft in the bushing area.

mm		in
47.536	MIN	1.8715
47.549	MAX	1.8720



If the drive shaft is **not** within specifications, it **must** be replaced.

Check the retaining rings for damage.

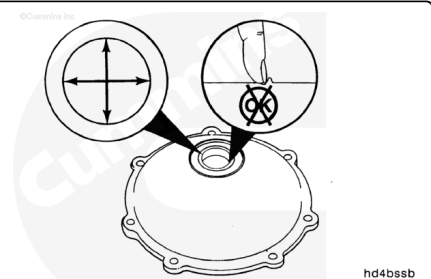
If the retaining ring damaged, it **must** be replaced.

Clean the hydraulic pump support bushing.

Inspect the hydraulic pump support bushing.

If a fingernail will catch on scratches or grooves, the bushing **must** be replaced.

Measure the bushing inside diameter.



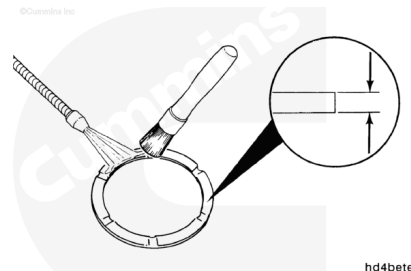
mm		in
47.60	MIN	1.874
47.68	MAX	1.877

If the bushing is **not** within specification it **must** be replaced.

Clean the thrust bearings.

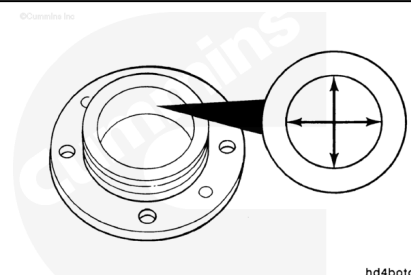
Measure the thrust bearing thickness.

mm		in
2.27	MIN	0.085
2.31	MAX	0.091



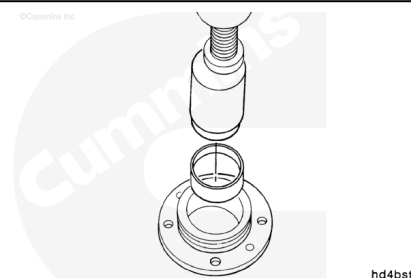
If the thrust bearing is **not** within specifications, it **must** be replaced.

Measure the hydraulic pump support bushing bore. Refer to Procedure 009-037 (/qs3/pubsys2/xml/en/procedures/20/20-009-037.html).



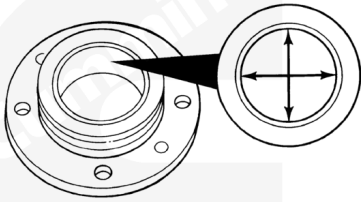
Assemble

Use the appropriate mandrel and press to install the bushing in the hydraulic pump support. Refer to Procedure 009-037 (/qs3/pubsys2/xml/en/procedures/20/20-009-037.html).



Measure the bushing inside diameter. Refer to Procedure 009-037 (/qs3/pubsys2/xml/en/procedures/20/20-009-037.html).

If the bushing is **not** within specifications, the support **must** be replaced.

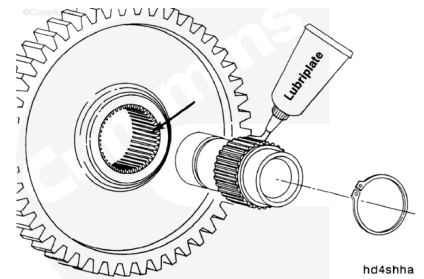


hd4bstg

Install the non-tapered snap ring onto the drive shaft.

Lubricate the external splines on the drive shaft and the internal splines on the gear with Lubriplate® 105, or equivalent.

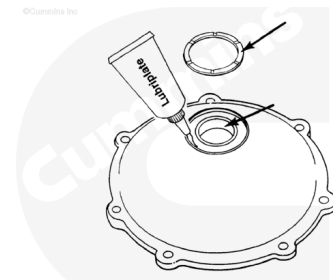
Align and install the gear onto the drive shaft.



hd4shha

Lubricate the hydraulic pump support bushing, counterbore, and one thrust bearing.

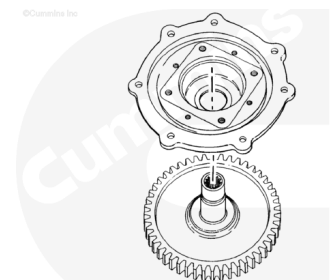
Place the thrust bearing, with the grooved side up, into the counterbore of the cone shaped end of the hydraulic pump support.



hd4beaa

Place the gear and shaft assembly on the end with the gear end down.

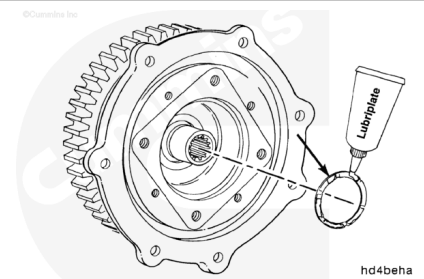
Align and install the hydraulic pump support cone end first onto the drive shaft and gear assembly.



hd4suaa

Lubricate a thrust bearing with Lubriplate® 105 or equivalent.

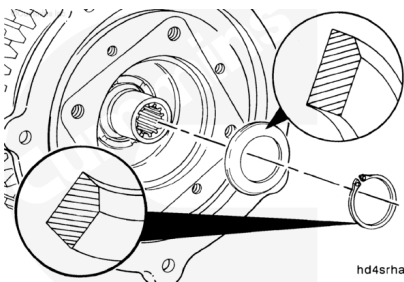
Install the thrust bearing into the counterbore of the hydraulic pump support with the grooved side up.



hd4beba

Install the bearing spacer over the drive shaft with the beveled side up.

Install the tapered retaining ring into the groove of the drive shaft with the beveled side up.

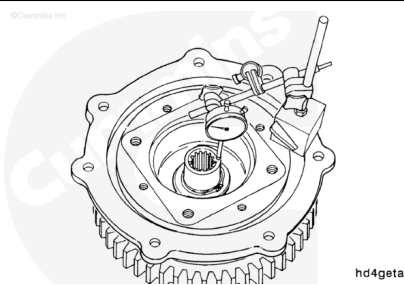


hd4srha

Measure the gear end clearance.

mm		in
0.10	MIN	0.004
0.51	MAX	0.020

If the end clearance is **not** within specifications, the thrust bearings **must** be replaced.



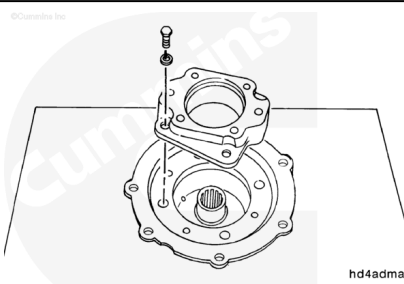
hd4geta

Install the hydraulic pump adapter or cover plate and gasket.

Install the capscrews.

Tighten capscrews.

Adapter	95 n.m	[70 ft-lb]
Cover Plate, SAE A Drive	40 n.m	[30 ft-lb]
Cover Plate, SAE B and C Drive	95 n.m	[70 ft-lb]

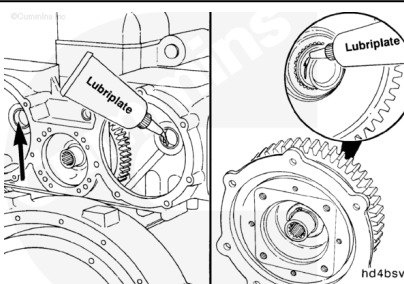


hd4adma

Install

Lubricate the outer hydraulic pump support bushings in the upper housing with Lubriplate® 105 or equivalent.

Lubricate the bushing surfaces on both shafts in the outer hydraulic pump support drive assemblies.



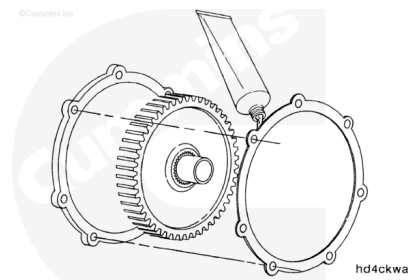
hd4bsva

Note : Some engines will **not** have an outer hydraulic pump support drive. These engines require a cover

plate and a non-splined shaft, but do not use a hydraulic gear.

Apply a thin coat of gasket adhesive to the gaskets.

Install the hydraulic support housing gaskets to both support mounting flanges on the outer hydraulic pump support drive assemblies.



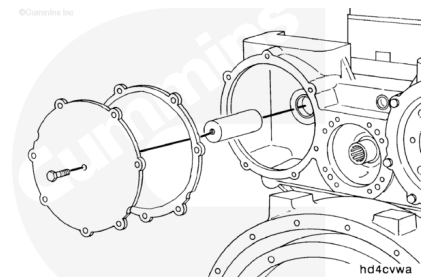
If using the outer hydraulic pump support cover plate, install the non-splined shaft onto the cover plate with a 3/8-16 x 1 1/4 inch capscrew.

Tighten the capscrew.

Torque Value: 40 n•m [30 ft-lb]

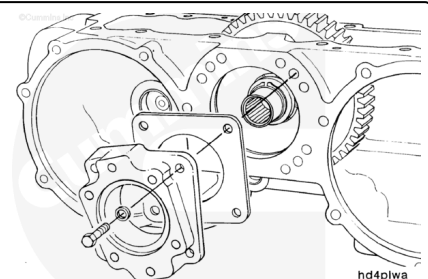
Install the cover plate assembly and capscrews.

Tighten the capscrews.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install three 7/16 - 14 x 4 guide studs.

Install one outer hydraulic pump support drive assembly.

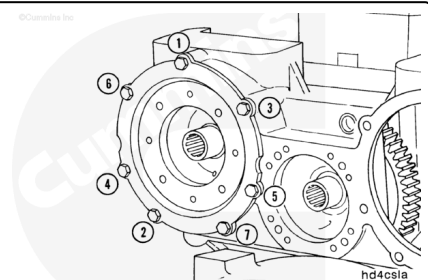
Install four capscrews.

Hand tighten the capscrews.

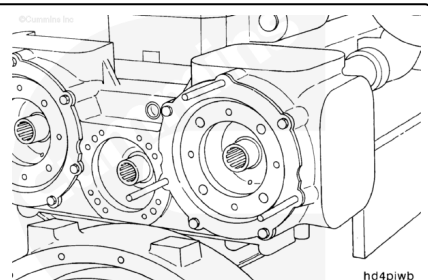
Install the remaining three capscrews.

Tighten the capscrews in sequence.

Torque Value: 70 n•m [50 ft-lb]



Use the same procedure to install the remaining outer hydraulic drive assembly.



Last Modified: 30-Jun-2006
